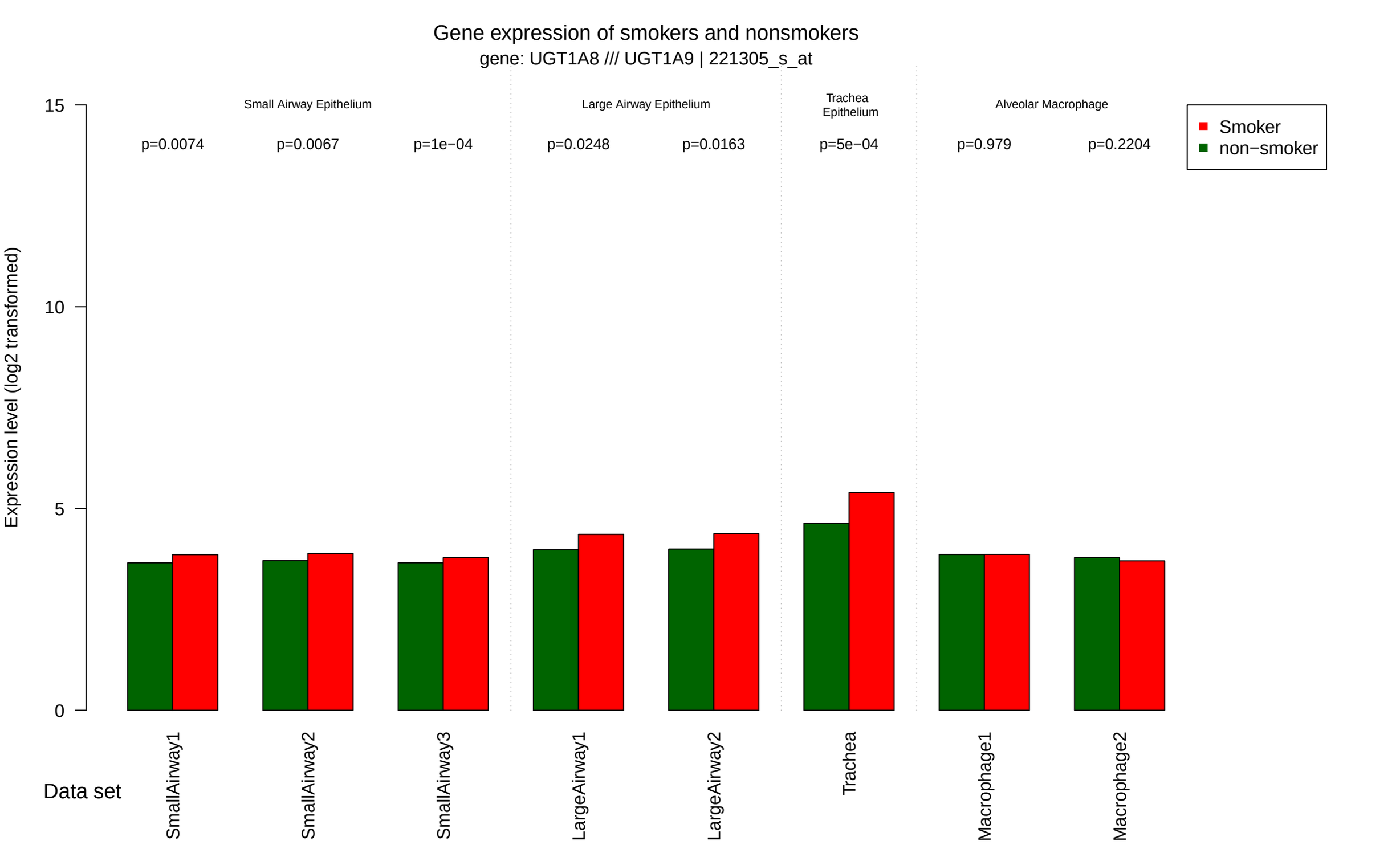
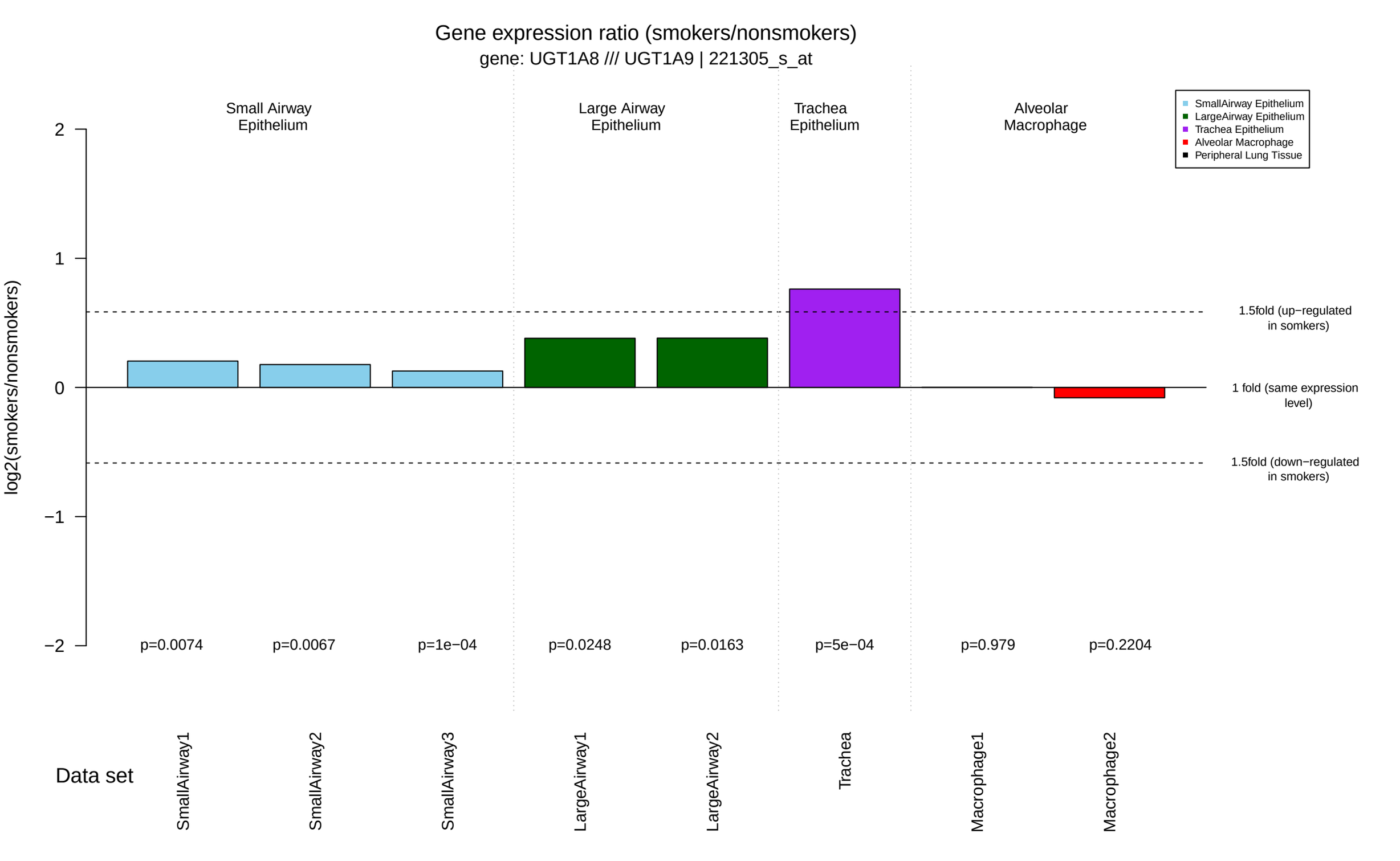
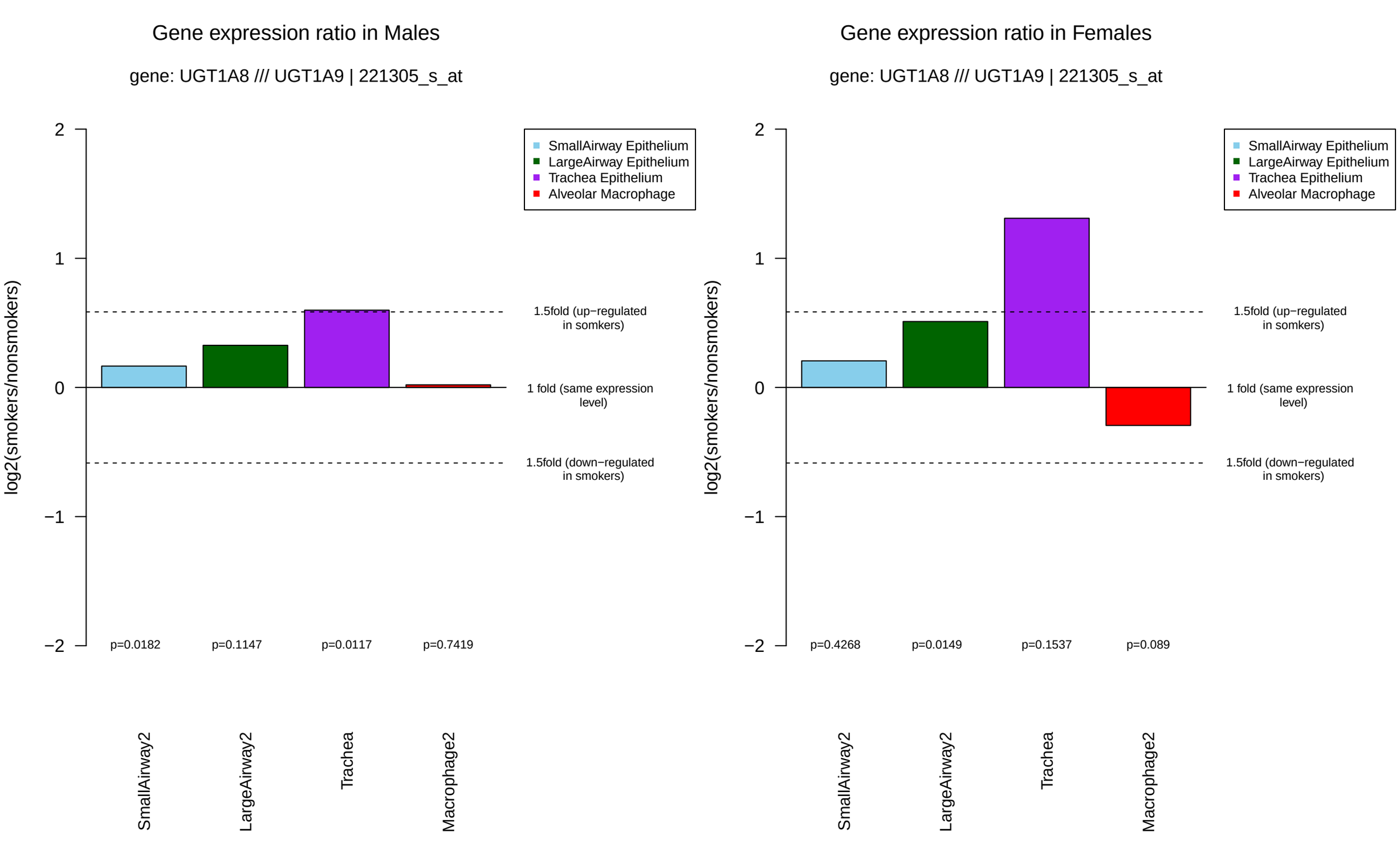
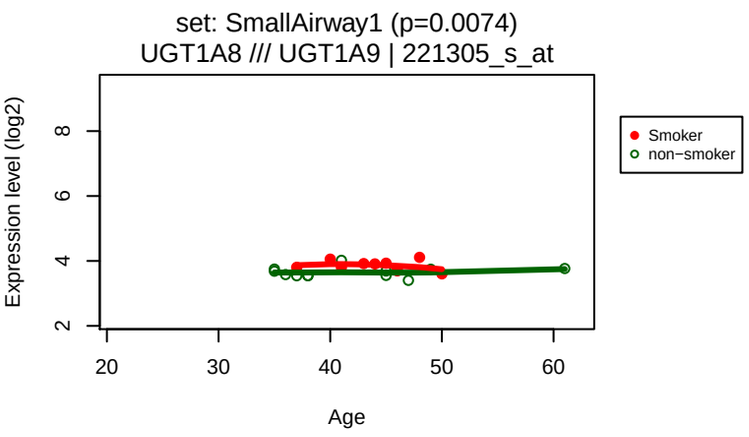
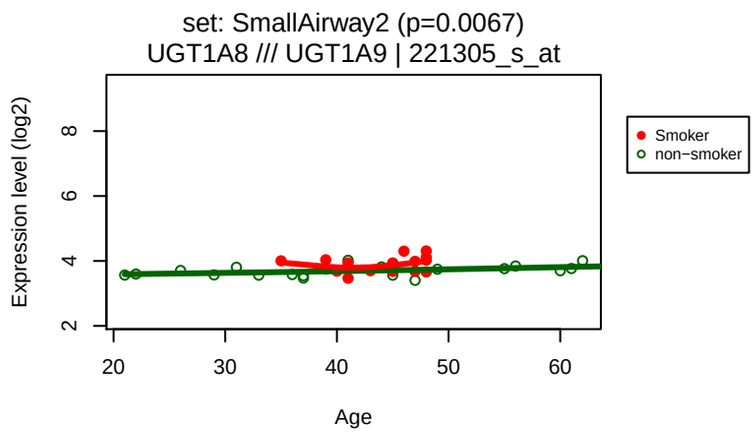
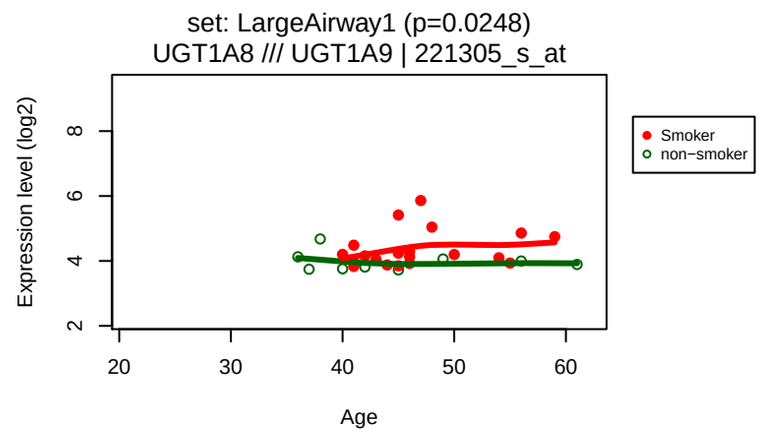

Gene expression of smokers and nonsmokers
gene: UGT1A8 /// UGT1A9 | 221305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0074
p=0.0067
p=1e−04
p=0.0248
p=0.0163
p=5e−04
p=0.979
p=0.2204
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

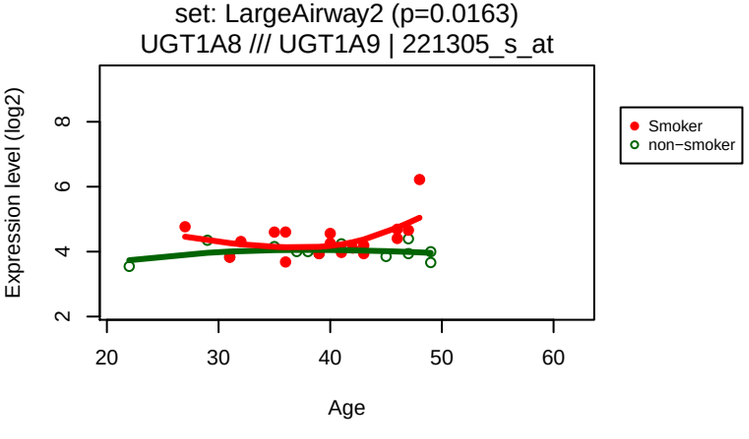

Gene expression ratio (smokers/nonsmokers)
gene: UGT1A8 /// UGT1A9 | 221305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.0067
p=1e−04
p=0.0248
p=0.0163
p=5e−04
p=0.979
p=0.2204
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

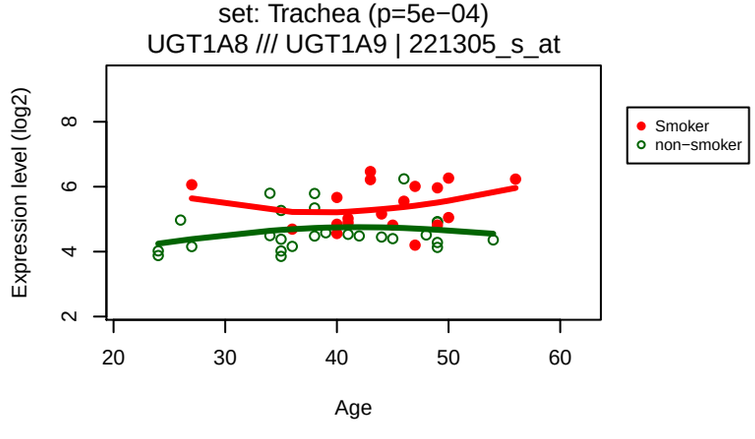

Gene expression ratio in Males
gene: UGT1A8 /// UGT1A9 | 221305_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0182
p=0.1147
p=0.0117
p=0.7419
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UGT1A8 /// UGT1A9 | 221305_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4268
p=0.0149
p=0.1537
p=0.089
SmallAirway2
LargeAirway2
Trachea
Macrophage2

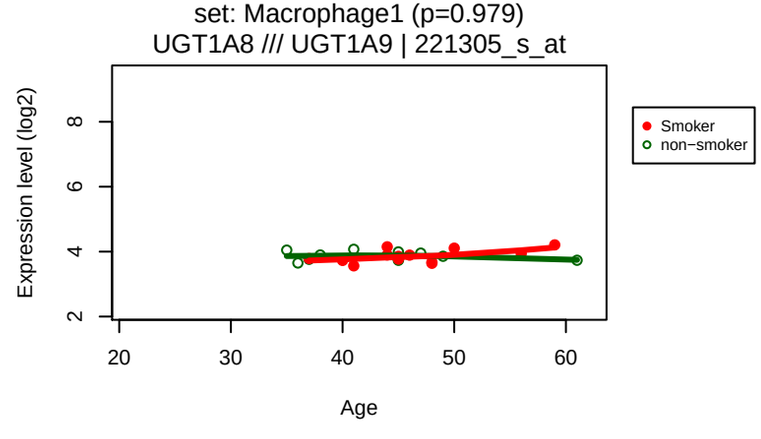

set: SmallAirway1 (p=0.0074)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker

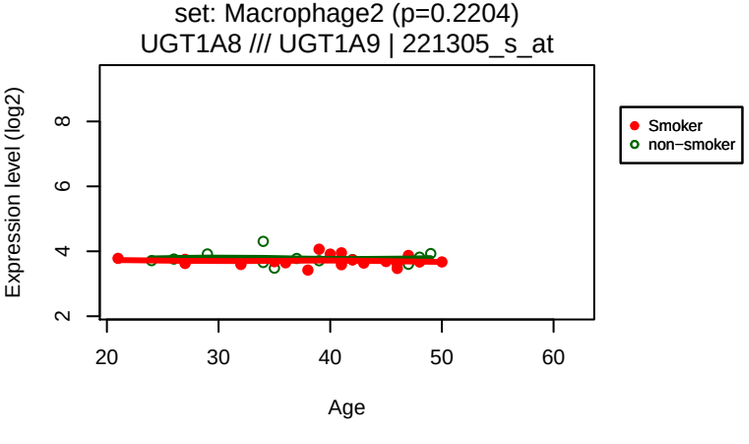

set: SmallAirway2 (p=0.0067)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker

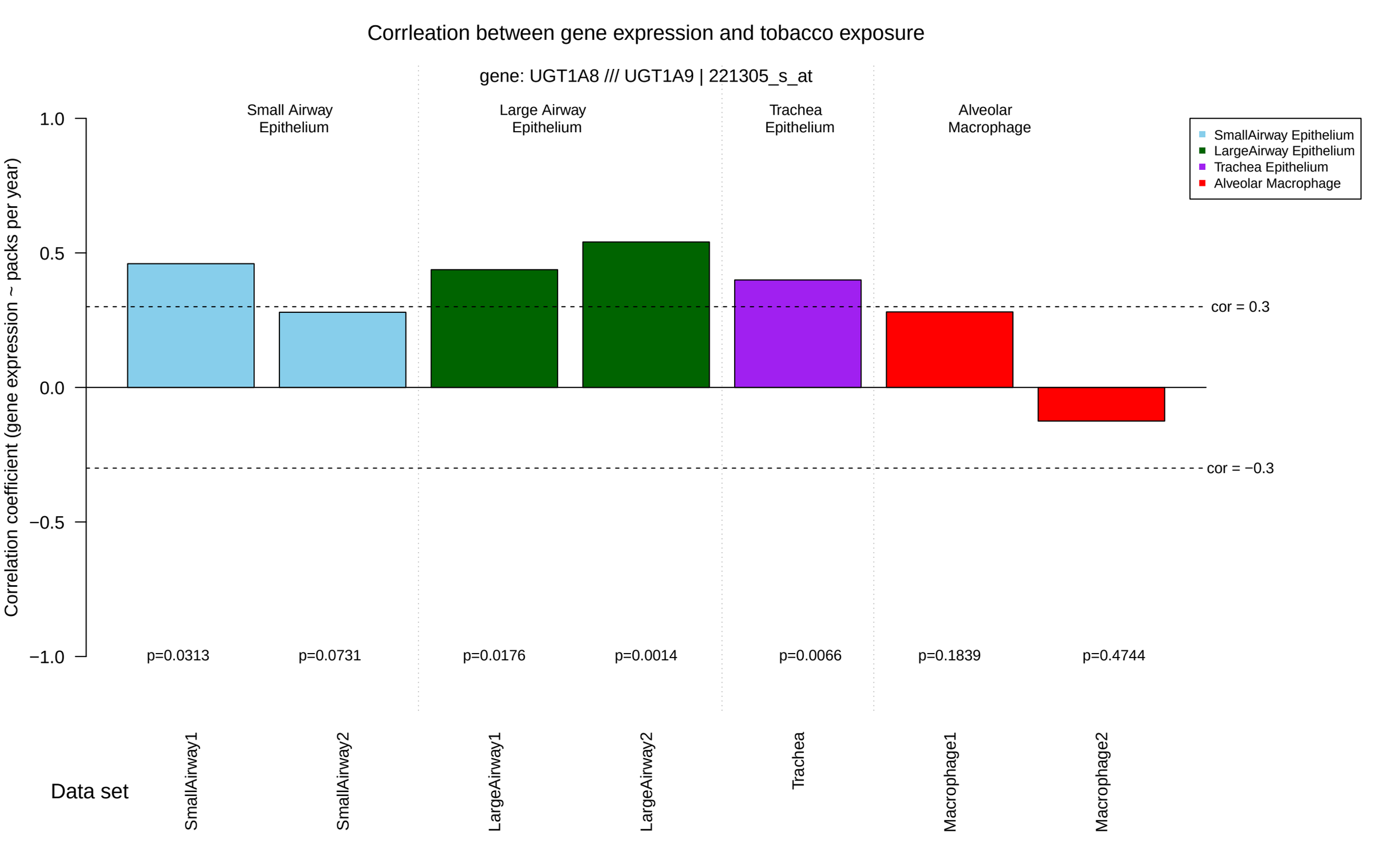

set: LargeAirway1 (p=0.0248)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0163)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.979)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2204)
UGT1A8 /// UGT1A9 | 221305_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UGT1A8 /// UGT1A9 | 221305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0313
p=0.0731
p=0.0176
p=0.0014
p=0.0066
p=0.1839
p=0.4744
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2